

Joanne Nguyen

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Professional Experience

Stealth Stage Startup | Hayward, CA

04/2025 - Present

Senior Research Associate, NGS Assay Development

- Established a robust assay pipeline where bead-conjugated samples consistently generated sequenceable NGS DNA libraries, confirming functional optimization of iterative testing and validation using qPCR
- Applied a design-of-experiments approach to optimize bead-based molecular assays, systematically varying reagents, concentrations, timings and conditions to maximize signal and minimize background
- Presented weekly experimental results, influencing decisions on assay optimization and product direction
- Developed traceable protocols and automated data-tracking tools (e.g. mastermix calculators, qPCR statistical analyses, reagent database look-up) to enhance reproducibility and streamline analysis

Mission Bio | South San Francisco, CA

02/2022 - 10/2024

Research Associate II, Pharma Assay Development

- Optimized single-cell data capture through Tapestry multiomics and improved NGS library workflows, resulting in higher-quality data collection, leading to more reliable outcomes in client projects targeting cancer therapeutics
- Led multiple high-priority projects simultaneously, meeting tight deadlines and improving experiment outcomes
- Collaborated with R&D project leads on projects goals, contributing to experiments which led to [6 publications](#)
- Integrated molecular techniques (bulk NGS, qPCR, gel electrophoresis, cell fixation and oligo conjugation) into existing protocols, enabling greater insight for troubleshooting process development and downstream analysis
- Implemented Javascript automation tools to streamline data management, reducing processing time by 50% through automated real-time updates for sample tracking, inventory management, and data filtering
- Built a Python regex-based tool to automate DNA tag pattern identification for statistical comparative models

Lifelabs Genetics | Toronto, ON

09/2020 - 01/2022

Site Lead

04/2021 - 01/2022

- Promoted within 7 months for demonstrating strong leadership in training and mentoring a 35-member team on high-complexity molecular assays, data analysis, and instrumentation for COVID clinical diagnostic analysis
- Developed and implemented CAP-compliant SOPs, ensuring assay consistency and regulatory adherence
- Analyzed qPCR data trends and presented findings to internal teams, identifying patterns in positivity rate
- Applied a design of experiments matrix to optimize positive control dilution, achieving a 60% cost reduction

Medical Laboratory Technician

09/2020 - 04/2021

- Ensured efficiency in high-throughput COVID RNA extraction, qPCR, mass spectrometry and analysis
- Coordinated qPCR and mass spectrometry workflows for 1000s of clinical samples a day with quick turnaround
- Led inventory organization and management by implementing Lean to reduce consumable waste
- Maintained, validated and calibrated qPCR analyzers and automated liquid handlers for optimal function

Other Experience: Research Assistant, Plant Genomics (University of Toronto | 08/2019 - 12/2019), Research Technician, Plant Genomics (University of Toronto | 05/2019 - 09/2019)

Technical Highlights

Molecular Biology: NGS & NGS Library Prep, DNA & RNA Extraction/Quantification, qPCR, Mass Spectrometry

Quality: SOP Writing, ISO 9001, ISO 15189, CAP, QC/QA, GLP, GDP, HIPAA/PHI, EHS

Software: Python, Javascript, SQL, GitHub, Confluence, JIRA, Salesforce Cloud Coach, LIMS, LIS

Education

University of Toronto | Honors BSc in Human Biology of Health & Disease, Minors in Physiology & Biology